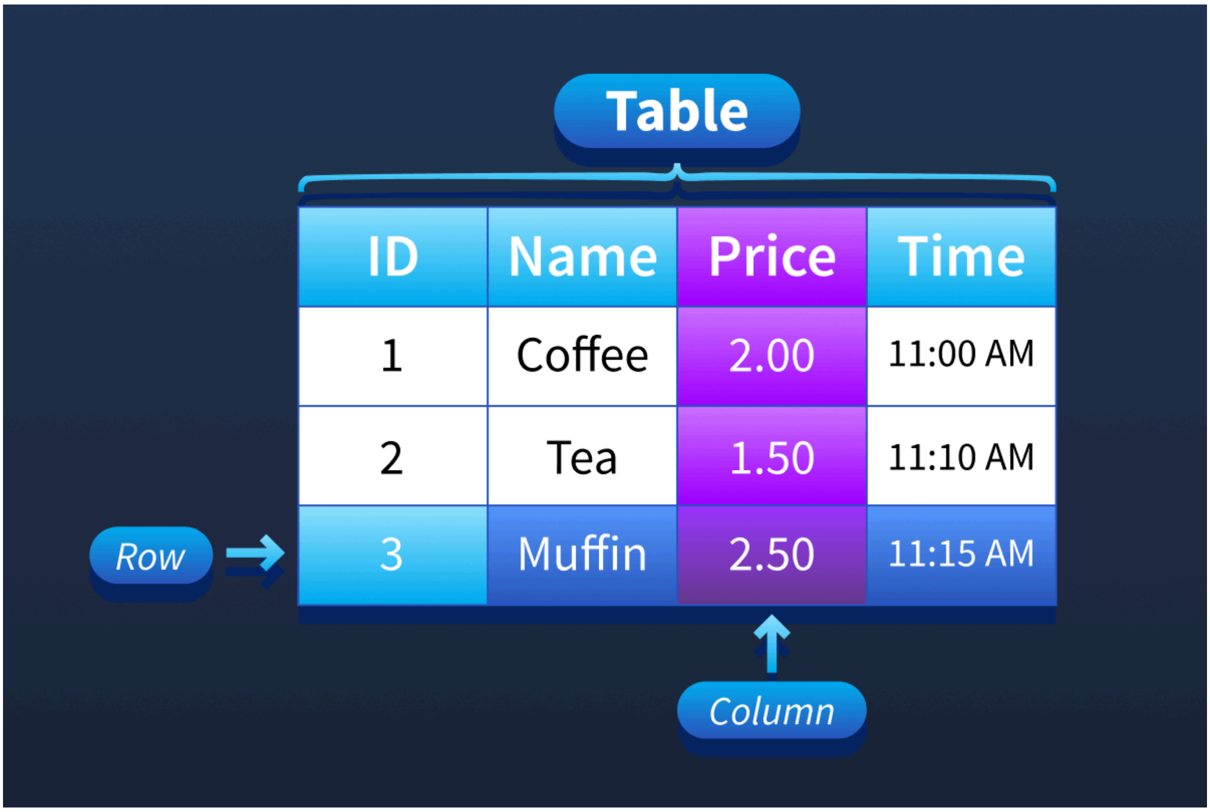
Path 1: Pre Security > Module 6: Software Basics > Room 5: Database SQL Basics

https://tryhackme.com/room/databasesqlbasics

Database SQL Basics
Learn the basics of databases and SQL by writing simple queries to retrieve and manage data.

>> Task 2: Understanding Tables, Rows, and Columns



- What Is a Database

Think of a **database** as a place where a computer stores information in an organised way.

You can think of a database as a digital notebook that never runs out of pages. Unlike a paper notebook, a database allows the computer to search, count, and sort information very quickly.

Even if the café has thousands of orders, a database can still answer questions in seconds.

Inside a database, information is stored in **tables**.

- Tables, Columns, and Rows

A **table** resembles a spreadsheet, where information is organised neatly into rows and columns.

- **Columns** are the titles at the top of the table. They describe the type of information stored.
- **Rows** go across the table. Each row contains one complete set of information.

For the café example:
- Columns might be: order number, drink, price, and time
- Each row is **one café order**

This means:
- One column stores one type of information, such as prices
- One row stores all the information about a single order

If the café sells ten drinks in one day, the table will contain ten rows. If one more customer places an order, one more row is added to the table.

If an order is removed, only that row disappears. The rest of the table stays the same.

- Asking Questions With SQL

SQL is a language used to ask questions of a database.

Instead of reading the table row by row, SQL lets the computer do the work for us.

For example, the café owner might ask:

- “Show me all orders.”
- “Show me only coffee orders.”
- “Show me the cheapest drink.”

These questions are called **queries**.

A query does not change the data. It only displays the requested information from the table.

Q1) Inside databases, what is the term for the "spreadsheets" that store the information?
Answer: Table

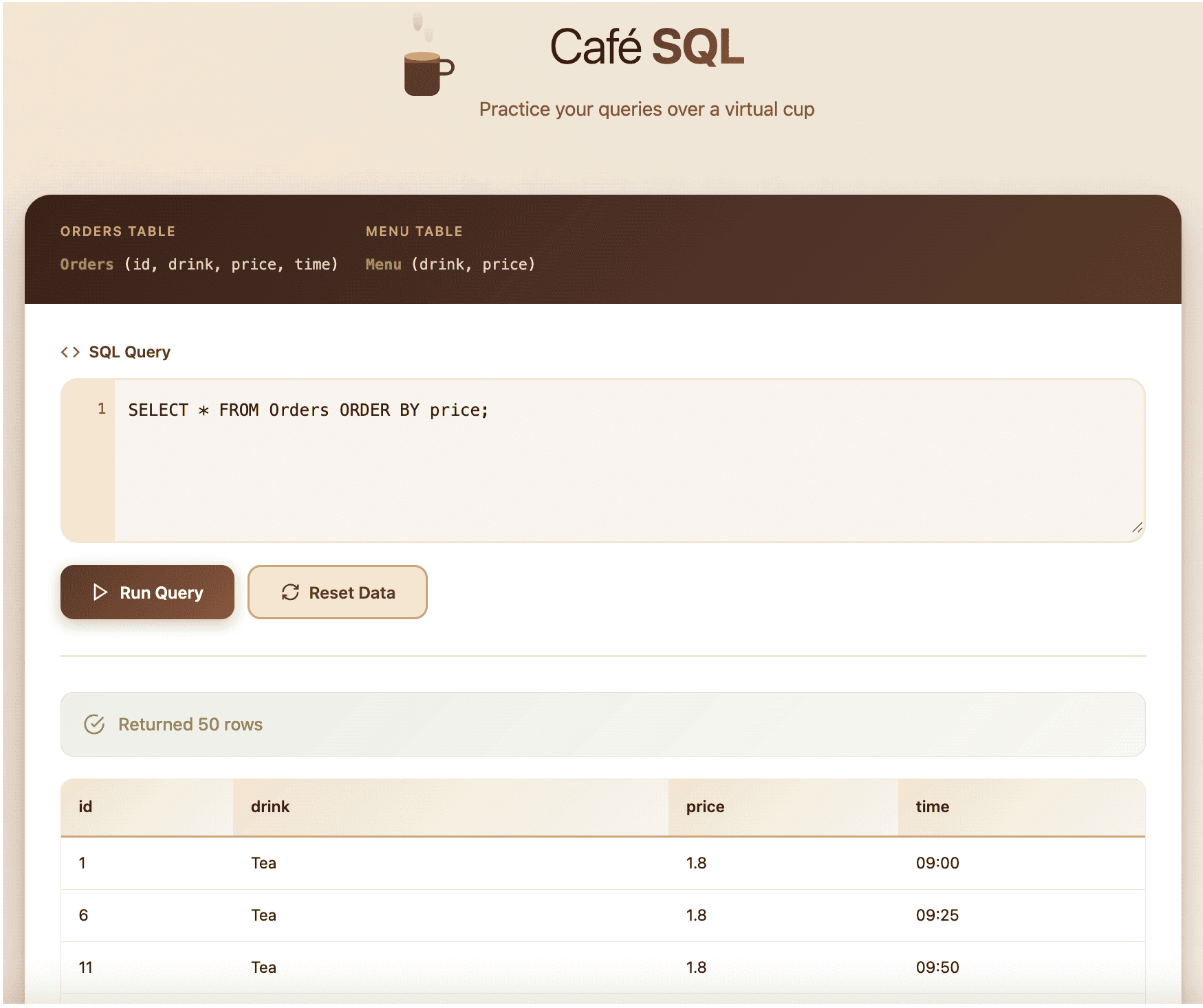
>> Task 3: Writing Your First SQL Query

- Step 1: View Everything in a Table (Select + From)

We start with the most basic query. When we use **SELECT ***, the * symbol means all columns. The word **FROM** tells the database which table to use.

Try this query:

SELECT * FROM Orders;



- Step 2: Show Only Specific Columns (Select Drink, Price)

Sometimes we do not need every column. We can choose specific columns by listing them after **SELECT**.

Try this query:

SELECT drink, price FROM Orders;

This will display only the **drink** and **price** columns.

- Step 3: Filter Results (Where)

The **WHERE** keyword filters rows. It keeps only rows that match a condition.

Try filtering by drink name:

SELECT * FROM Orders WHERE drink = 'Coffee';

If the database contains coffee orders, you will now see only those rows.

Hint: If you are not sure which drink names exist, run:

SELECT * FROM Menu;

- Step 4: Sort Results (Order By)

The **ORDER BY** keyword sorts results by a column. By default, results are sorted in ascending order (lowest to highest).

Try sorting orders by price (lowest first):

SELECT * FROM Orders ORDER BY price;

To sort in reverse order (highest to lowest), add **DESC**.

Try sorting orders by price (highest first):

SELECT * FROM Orders ORDER BY price DESC;

- Step 5: Combine Filtering + Sorting

Most real queries combine parts together. Here, we filter to keep only one drink type and then sort by price.

Try this query:

SELECT * FROM Orders WHERE drink = 'Coffee' ORDER BY price DESC;

Q1) When you showed all orders, how many rows were returned?
Answer: 50

Q2) When you sorted orders by price from cheapest to most expensive, which drink appeared first?
Answer: Tea

Q3) When you sorted the menu by price from most expensive to cheapest, which drink appeared first?
Answer: Latte

>> Task 4: Conclusion

🔹 SQL Queries Learned in This Room

SELECT – choose what data to display

FROM – choose where the data comes from

WHERE – filter records based on a condition

ORDER BY – sort results